SUNYANI TECHNICAL UNIVERSITY

FACULTY OF ENGINEERING

MECHANICAL ENGINEERING DEPARTMENT

A REPORT OF INDUSTRIAL ATTACHMENT

AT

SUNYANI TECHNICAL UNIVERSITY AVIC LABORITORY

PRESENTED BY

GBIREH RANOUS

INDEX NUMBER: STUBTECH221619

(TOP-UP, BTECH LEVEL 200 – "PRODUCTION" OPTION)

DECLARATION

I **GBIREH RANOUS,** do declare to the best of my knowledge, that this report is originally mine; it is a true record of the work I undertook during my industrial internship training at **SUNYANI TECHNICAL UNIVERSITY-AVIC LABORITORY** and has never been submitted for the award of the Bachelor of Technology in Mechanical Engineering or any academic qualification to the Sunyani Technical University or any other University or Academic Institution of higher learning.

Signature: …… Date: ……..

Name: GBIREH RANOUS

Index Number: STUBTECH221619

CERTIFICATION

I hereby certify that, this industrial internship training report was submitted to Sunyani Technical University, Mechanical Engineering Department.

Signature: …… **Date:** …….

Dr. Samuel Adu-Gyamfi (PhD)

Head of Department.

DEDICATION

This work is fully dedicated to my supervisors and fellow interns, for their greatest support through my study at the University.

ACKNOLEDGEMENTS

I thank the almighty God for His protection, guidance and his unconditional love for making the training a success. It is entirely by His grace that I have reached this far in the academic journey praise glory and honor be to Him.

It would be impossible to acknowledge all the various persons whose positive influence to my life at various stage have shaped me to who I am today.

I extend my heartfelt gratitude to the management of **AVIC LABORITOR** for permitting me the valuable opportunity to train at their organization.

I would like to thank master Daniel Boateng Duah and Mr. Abubakar Yakubu for their invaluable contribution when I was under their custody as student learner. Both were my supervisors, they welcome me well and drop an advice that, I should be submissive in all my way and by the time I would be done with my internship, I will know what I desire to achieve.

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INTRODUCTION

A SUMMARY OF SUNYANI TECHNICAL UNIVERSITY-AVIC LABORITORY AND ACTIVITIES DURING THE ATTACHMENT

Sunyani Technical University present a very interesting history from its inception in the late 1960 to date as it continues to carve a niche for itself in its contribution towards meeting the human resource needs of the country. It started in 1967 as Sunyani Technical Institute and offered a great opportunity for middle school leavers and later junior secondary school graduates to undergo hands-on training in craft/inter -mediate Block-laying and concreting, carpentry and joinery, furniture craft as well as painting and decoration. Other programs offered include electrical installation, Motor Vehicle Mechanics, Mechanical Engineering Craft, etc. In 2016, the polytechnic was converted to a technical university following the promulgation of the Technical Universities Act, 2016 (Act 922) as amended. The Act 922 mandates the University to provider higher education in Engineering, Science and Technology based disciplines, TVET, applied arts, and related discipline and award degrees including, honorary degrees, diplomas, certificate and honor qualifications agreed upon by the governing council. Currently the university offers a number of Bachelor of Technology Degree programs, Higher National Diploma and Non-Tertiary programs in TVET, technology based and science related disciplines. Preparations are also underway to obtain the requisite accreditation for the running of Master of Technology and Doctor of Technology in the near future.

SUNYANI TECHNICAL UNIVERSITY-AVIC LABORITORY was inaugurated in August, 2022. The AVIC laboratory encompasses various department:

- Mechanical Engineering department
- Civil Engineering department
- Materials and Welding department
- Electrical Engineering department

I was privileged to undertake my industrial internship in the Mechanical Engineering department-production section.

DESRIPTION OF WORK DONE AT AVIC LABORATORY

MACHINE OPERATION/ MACHINING

In the manufactory industries various branches are been instituted where machine operation/ machining (machine workshop) happens to be one of the important branches. Machine operation/machining is the process of using any machine in the machine shop being lathe machine, milling machine, shaping machine, etc. to produce a component to the required shape and dimensions. That is, to be in accordance with the design and the working drawing of that component.

Haven said earlier on that AVIC laboratory consist of four workshops which are;

- Mechanical Engineering department
- Civil Engineering department
- Materials and Welding department
- Electrical Engineering department

. The emphases are on the mechanical workshop (production shop) because the attachment was based on that.

At AVIC laboratory, various kinds of machines are installed in the workshop. During the attachment period, the following machines were some of the machines that we had the chance to use in the practices of our practical training which are as follows; the lathe machine, shaping machine, drilling, grinding machine, CNC lathe machine, CNC milling machine, etc.

OPERATIONAL ACTIVITIES THAT WERE BEEN DONE ON THE MACHINES.

The lathe machines.

- Step turning: is the process of machining a work piece (hollow or solid shaft) on the lathe machine in other to achieve different external diameters on the same shaft.
- Parallel turning: is the process of machining a work piece (hollow or solid shaft) on the machine in other to reduce the external diameter of the shaft uniformly.
- Facing: is the process of squaring the ends of a round work piece (hollow or solid) on the lathe machine. To this operation, the cutting tool in the tool post on the compound slide is set at an angle not equal or more than 90 the appropriate angle is 45.

- Thread cutting: a thread is uniform helical shape that is been cut on the external part of a shaft or internal part of a hollow shaft. Cutting thread on the lathe machine requires the following; maximum and minimum diameter of the work piece, the pitch of the thread should be known and a thread cutting tool.
- Cutting of a v- groove on a pulley to receive a v- belt.

The shaping machines

- Cutting of keyways both external and internal keyways on shafts and in pulleys.

The grinding machines

- Grinding of cutting tools and drill bits to the required angle and shape in other for them to cut effectively when machining or drilling.

Power hacksaw

- Cutting of large size sections of metals, such, steel. Cutting diameters of more that 10/15mm is very hard work with a normal hand held-hacksaw.

PRACTICAL ACHIEVEMENTS

- How to set work piece in a four-jaw chuck.
- How to set work piece at the center height of the lathe machine.
- How to set and select gears when turning or cutting thread on the lathe machine.
- How to grind cutting tool at a required angle and shape.
- How to rivet.

IMPACT OF THE ATTACHMENT

The attachment has positively helped students in the following areas.

- Proper time management.
- Safety in the workshop.
- Machine operation/machining.

PROPER TIME MANAGEMENT

Time is the space between two events. Time is the most important factor in engineering if we must improve or progress in the field of engineering. For that matter, we need to obey and follow time diligently in our field of study and profession. Researches have proved that in other for one to meet deadlines he/she must stick to the time assigned to each of the activities. This does not help one to meet deadlines alone but also shows how competent and reliable we are in our field of study and profession. When we follow time properly, at long round productivity will increase and hence poverty will also decrease.

Based on these facts we were been admonished to obey and follow the time assigned for reporting to work and the time we should close. This was done by the training director of the organization on our first day that we reported for our attachment. With his advice some

of us (students) did obey till the end of our attachment period. The working hours of the organization are as follows 8:00 am to 5:00 pm from every Monday to Friday.

Haven passed through these directives, especially we those who were present and punctual at work. It has positively helped us to be ready to readjust ourselves and move forward in case we encounter any difficult situation (concerning time) in our field of study or profession.

SAFETY IN THE WORKSHOP

Safety can be described as the necessary steps or rules that are put in place in other to reduced and prevent accidents in one's life. In life if we must live safe and out of avoidable dangers, we set up safety precautions and practice them without compromising. Therefore, safety is considered as a major concern in the field of engineering such as mechanical engineering, electrical/electronic engineering, computer engineering, civil engineering, aerospace engineering etc. and the field of medicine. Safety in general does not only focus on the personnel alone but also the machines and tools used by the personnel.

The main reason why safety must be observed in the workshop is as follows.

- To ensure safe working environment for both employees and customers. Which result in the reduction and the prevention of avoidable accidents in and around the workshop?
- To prolong the life of machines, tools and equipment. Which result in maintaining the efficiency of the machines and hence productivity will be increased.
- To improve on accuracy and precision in the line of servicing and production of products.

In view of all these we never joked (compromised) in obeying and practicing the safety precaution that governs the organization and what we were been taught in school. Our obedience to the safety precaution that governs the organization and what we learned back at school have helped us to stay out of danger during and after our attachment period.

The table below shows some of the safety precaution and the dangers associated to each of them if not properly followed or obeyed in the machine workshop.

SAFETY PRECAUTIONS	**DANGERS/EFFECTS WHEN NOT FOLLOWED**
Before one should operate on any machine, he/she should wear the prescribed safety cloths.	Failure to do that can cause injury to the following parts of the body; the fingers, eyes, toes and the head.
Constantly check, clean, service and repair all minor problems on the machines before and after using them.	Failure to do that can result low efficiency and failure of part of the machine or total breakdown of the machine
On the lathe machine, when machining ensured that the work piece is set the center height of the machine.	When work piece is set above or below the center height. The cutting tool either robs or digs into the work piece. Hence it causes high frictional noise and the rake angle of the cutting tool is either decreased or increased depending the situation at hand.
Measuring instruments should be regularly and constantly calibrated.	Failure to do that leads to incorrect measurements which affects accuracy and precision of items been produced.

CONCULSION

Despise the challengers that students do encounter during the attachment period, it has always been helping students to become practical oriented in their various fields of studies. To summary, the attachment has greatly brought improvement in the following areas;

- Punctuality and decency in the field of work.
- Clear understanding of some of the theory knowledge that we learned in school.
- The readiness to work with different kinds of people without hindrances.
- The chance of meeting and learning of new things that were never experienced.

GENERAL OBSERVATION ABOUT THE ORGANIZATION

The observations about the organization are in line with the following;

- Conducive environment which gives way for smooth working.
- Rules are strictly followed.

Remarks: industrial attachment is the main way of linking academia and the industry.

APPENDIXES

WORKSHOP

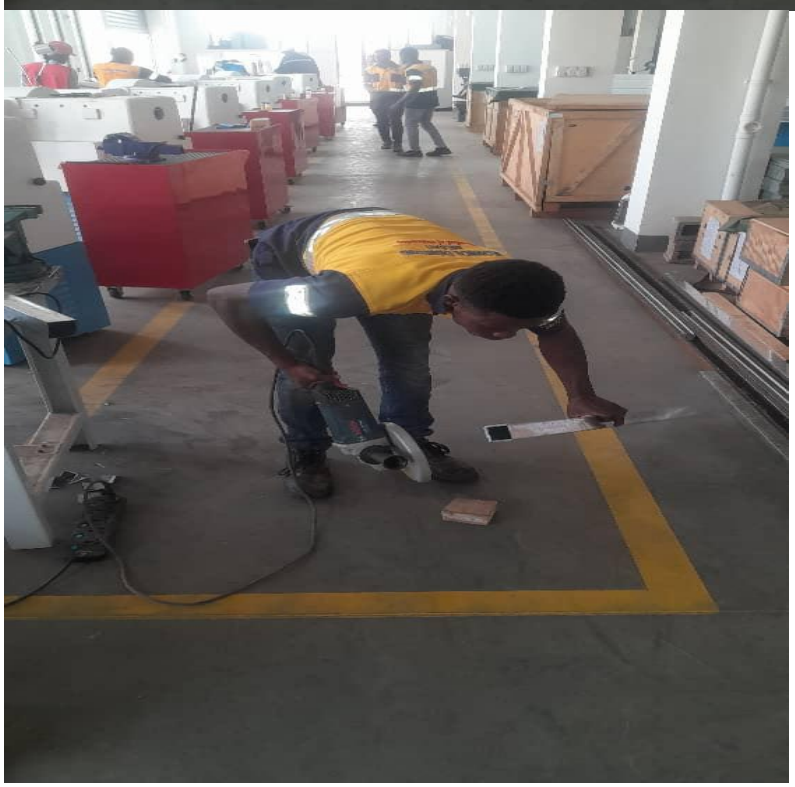
AVIC INTL
Vocational Training

AVIC INTL
Vocational Training

AVIC INTL
Vocational Training
TURNAC
CA141X1000

VMCL 850T
AVIC INTL
Vocational Training

PICTURE OF ACTIVITIES DURING THE ATTACHMENT



FORM A

SUNYANI TECHNICAL UNIVERSITY

INDUSTRIAL ATTACHMENT- ASSESSMENT FORM

NAME OF STUDENT Gabineh Ronald ..... INDEX NUMBER STUBTECH221619
COURSE Mechanical Engineering ..... CLASS/YEAR 200 (production)
DURATION OF ATTACHMENT Five months .....
NAME AND ADDRESS OF HOST ORGANISATION Sunyani Technical University AVIC laboritory Box 206 .....
TEL: NO 0241226743 ..... LOCATION Sunyani .....
JOB ASSIGNED TO STUDENT 1. Taper turning at shaft
2. How to cutt keyway internal and external
3. I was assigned to drill ten of 6mm hole through a plate.

SCORING 5=EXCELLENT
4=GOOD
3=AVERAGE
2=BELOW AVERAGE
1=POOR 0=ABSENT

A. SPECIAL SKILLS (Skills related to work Assigned) List them.	CIRCLE TO INDICATE PERFORMANCE	COMMENTS IF ANY
1. MILLING MACHINING	0 1 2 (3) 4 5	
2. LATHE MACHINING	0 1 2 3 (4) 5	
3. CNC MACHINING	0 1 2 (3) 4 5	
4. SML FACE & CYLINDRICAL MACHINING	0 1 2 (3) 4 5	
SUBTOTAL SCORE		
B. GENERAL EMPLOYABILITY SKILLS		COMMENTS IF ANY
5 Ability to complete work on Schedule	0 1 2 3 (4) 5	
6 Ability to follow Instructions Carefully	0 1 2 3 (4) 5	
7 Ability to take initiatives	0 1 2 (3) 4 5	
8 Ability to work without Supervision.	0 1 2 3 (4) 5	
9 Ability to work with other Staff.	0 1 2 (3) 4 5	
10 Adherence to Organization's Rules and Regulations.	0 1 2 (3) 4 5	

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11 Adherence to safety and Environmental Rules	0 1 2 3 4 5	
12 Resourcefulness	0 1 2 3 4 5	
SUBTOTAL SCORE		
C.ATTITUDE TO WORK		COMMENTS IF ANY
13 Attendance to Work	0 1 2 3 4 5	
14 Punctuality	0 1 2 3 4 5	
15 Desire to Work	0 1 2 3 4 5	
16 Willingness to accept additional responsibility (s)	0 1 2 3 4 5	
SUBTOTAL SCORE		
D.HUMAN RELATIONS		COMMENTS IF ANY
17 Relationship with Subordinates	0 1 2 3 4 5	
18 Relationship with colleagues	0 1 2 3 4 5	
19 Relationship with Superiors	0 1 2 3 4 5	
20 Emotional control(Ability To control emotions when Provoked)	0 1 2 3 4 5	
SUBTOTAL SCORE		

TOTAL SCORE.....70.....

GENERAL REMARKS...RESPECTFUL AND EAGER TO LEARN.....

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NAME OF SUPERVISOR ENGR. ALEX AMPOFO TEL. NO. 0246290291

SIGNATURE AND OFFICIAL STAMP.....

LECTURER
SUNYANI TECHNICAL UNIVERSITY
MECHANICAL ENGINEERING DEPT.
SUNYANI

NOTE: THE FORM SHOULD BE SENT TO THE LIAISON OFFICER ONE MONTH AFTER THE ATTACHMENT PERIOD.

REGISTRAR
INDUSTRIAL LIAISON OFFICE
SUNYANI TECHNICAL UNIVERSITY
P.O. BOX 24 SUNYANI

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